

Reference excerpt 01-025

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$c \gg w$. [Womersley(1955c)] The solution of Eq.1 and the complete procedure to calculate it are explained in detail in [Sisini et al.(2015b)]. However the solution is then defined up to the multiplicative constant c and up to the additive constant w_0 . As a consequence, the instantaneous velocity trace, calculated in this way, is proportional but not equal to the actual blood velocity. Pressure waves propagation The pressure is transmitted along the direction AR-IJV according to the following equation $p(z) = p(t) \pm \frac{z}{c}$ (2) For each cardiac cycle, the pressure in the RA reaches its maximum value between t_P and t_Q , where t_P indicates the ECG P wave and t_Q indicates the wave Q. In this note, for the sake of simplicity, it is assumed that the pressure in the RA is maximum in correspondence of the instant t_{PQ} which is the average between t_P and t_Q . The relationship between the pressure in the RA and the pressure in the point G at the same instant of time is given by: $p(t + \frac{IG}{c}, z_G) = p(t, z_{RA})$ (3) Where z_G is the coordinate of the US scanning point and z_{RA} is the RA coordinate. The pressure is, at the same instant t_{PQ} , maximum in the RA ($p(t_{PQ}, z_{RA})$), and equal to $p(t_{PQ} - \frac{l}{c}, z_{RA})$ at the insonation point G. The qualitative JVP trace of 2 Figure 2: The figure a) shows the JVP trace of pressure in the IJV, measured in G, together